

ENGINEERING BULLETIN

PRODUCT IMPROVEMENT

Improved Oil Recovery Line for Flooded Screw Chillers

➤ Introduction & Program Overview

KCPL is pleased to inform an improved oil recovery line for water or air cooled flooded screw chillers. This product improvement is aimed at enhancing chiller performance by improving oil recovery from the evaporator during both full-load and part-load operating conditions, while simplifying the system design by eliminating the solenoid valve/s in the oil recovery line.

Improved Design

Dual Compressor Chiller: In legacy design, when the chiller operated at part load (with only one compressor active), the recovery line for the idle compressor circuit remained inactive.

In improved design, the solenoid valves are eliminated. Both oil recovery lines remain fully operational regardless of whether one or both compressors are running. Recovery lines are connected to the suction lines of both compressors with additional tee and angle valves as shown in the Figure 1 below.

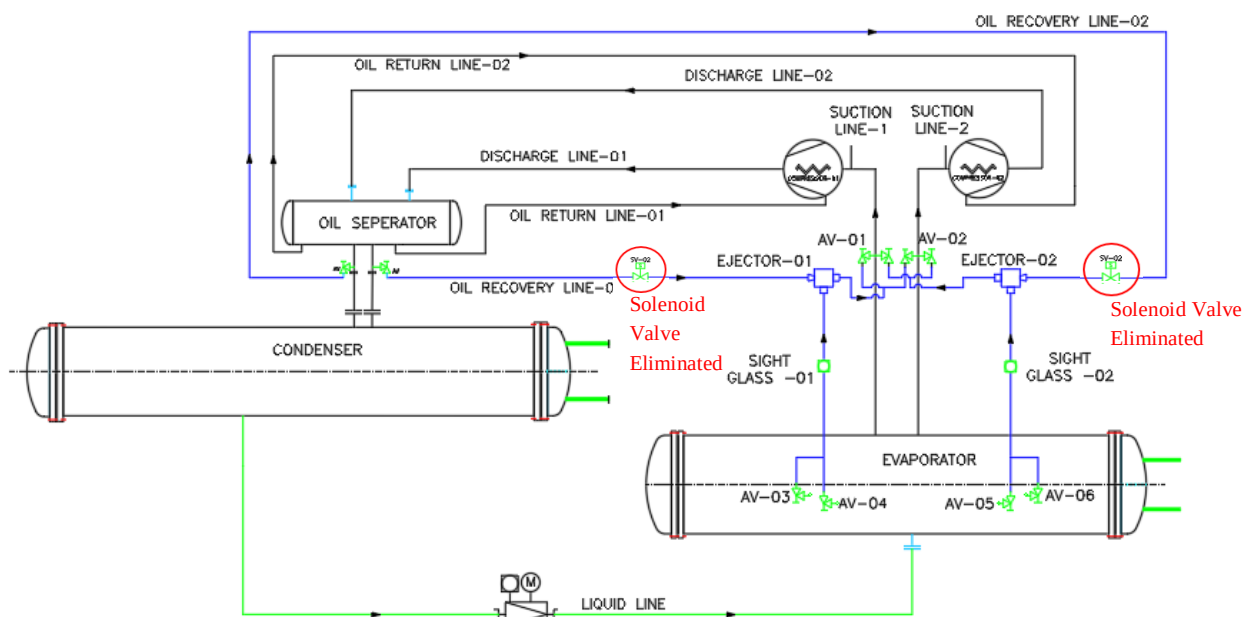


Figure 1: Improved Oil Recovery Line- Dual Compressor

Single Compressor Chiller: In improved design solenoid valves are eliminated. Oil recovery will be continuously on as shown in Figure 2.

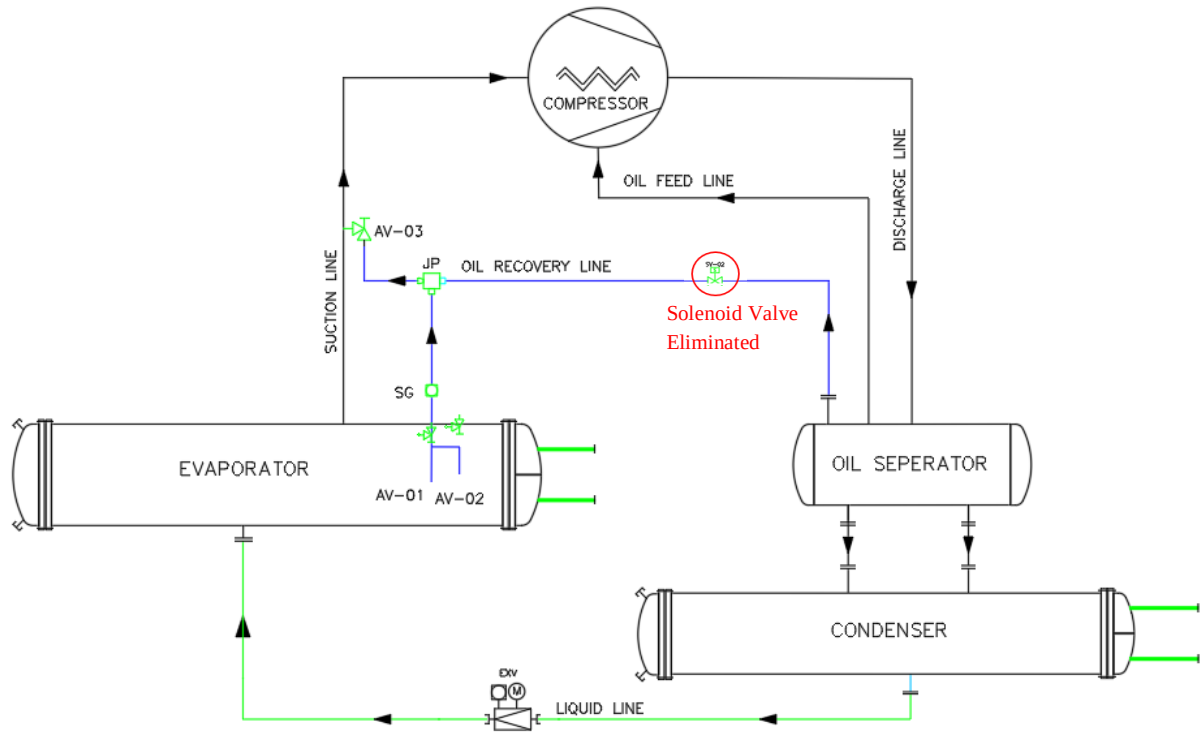


Figure 2: Improved Oil Recovery Line- Single Compressor Chiller

Benefits

- Solenoid valves eliminated.
- The oil recovery remains active at all loads.